

Design a Retrieval Strategy for a RAG System that Handles Both Structured Data (Knowledge Graphs) and Unstructured Data (Text Documents)

Overview

In many enterprise systems, information exists in both structured and unstructured formats.

- Structured data includes knowledge graphs, databases, and relational data that store information in a structured form such as entities and relationships.
- Unstructured data includes documents such as reports, emails, PDFs, manuals, and wiki pages.

A RAG system that needs to work effectively in such environments must implement a hybrid retrieval strategy that can retrieve relevant information from both types of data simultaneously.

The goal is to combine semantic search, keyword search, and graph-based retrieval to obtain the most relevant knowledge before sending it to the language model.

Types of Data in the System

Structured Data (Knowledge Graphs)

Structured knowledge is stored in the form of entities and relationships.

Example:

- (Employee) → works_on → (Project)
- (Service) → depends_on → (Database)
- (Customer) → purchased → (Product)

This type of data is stored in graph databases such as:

- Neo4j

- Amazon Neptune
- TigerGraph

Knowledge graphs help capture relationships between entities that may not appear directly in documents.

Unstructured Data (Text Documents)

Unstructured data consists of text-based documents such as:

- company manuals
- policy documents
- support tickets
- emails
- research papers
- wiki pages

These documents must be processed using semantic embeddings so that the system can retrieve them using vector similarity.

Hybrid Retrieval Strategy

A robust RAG system should use multiple retrieval methods simultaneously. The most effective strategy combines the following three approaches:

1. Semantic Retrieval (Vector Search)
2. Keyword-Based Retrieval
3. Graph-Based Retrieval

Semantic Retrieval for Unstructured Data

Semantic retrieval is used to search unstructured documents based on meaning rather than exact keywords.

Process

1. Documents are divided into smaller chunks.
2. Each chunk is converted into an embedding vector.
3. The vectors are stored in a vector database.
4. The user query is converted into an embedding.
5. Similar chunks are retrieved using vector similarity.

Vector Databases

Examples include:

- FAISS
- Pinecone
- Milvus
- Weaviate

Advantages

- Handles natural language queries
- Captures semantic meaning
- Works well with long documents

Keyword-Based Retrieval

Keyword retrieval helps find documents that contain exact words or identifiers.

This is useful when queries contain:

- product IDs
- error codes
- technical terms
- specific names

Implementation

Keyword retrieval is typically implemented using:

- BM25
- Elasticsearch
- Lucene

Example

Query:

"Error code ERR5042 in payment service"

Keyword search helps retrieve documents containing ERR5042, which semantic search might miss.

Graph-Based Retrieval for Structured Data

Structured knowledge stored in knowledge graphs can be retrieved using graph traversal.

Process

1. Identify entities in the user query using Named Entity Recognition (NER).
2. Locate the corresponding nodes in the knowledge graph.
3. Traverse connected relationships to retrieve relevant information.

Example

Query:

"Which services depend on PostgreSQL?"

Graph traversal:

PostgreSQL → used_by → Services

The system can retrieve all services connected to that database node.

Benefits of Graph Retrieval

- Captures relationships between entities
- Enables multi-hop reasoning
- Improves factual accuracy

Combining Results from Multiple Retrieval Sources

Since results come from different sources, they must be merged and ranked before sending them to the LLM.

Steps

1. Retrieve candidate documents from:
 - vector search
 - keyword search
 - knowledge graph traversal

2. Combine the retrieved results.
3. Apply ranking algorithms such as:
 - Reciprocal Rank Fusion (RRF)
 - Cross-encoder re-ranking
4. Select the top-k most relevant results.

These results form the final context for the language model.

Final Retrieval Workflow

The complete retrieval strategy works as follows:

1. User submits a query.
2. Query is analyzed to extract entities and meaning.
3. The query is processed using three parallel retrieval methods:
 - vector search on document embeddings
 - keyword search on document indexes
 - graph traversal on the knowledge graph
4. Results from all retrieval sources are combined.
5. A re-ranking model selects the most relevant information.
6. The selected data is passed to the LLM to generate the final response.

Advantages of This Hybrid Retrieval Strategy

Using both structured and unstructured retrieval provides several benefits:

- Improves answer accuracy
- Enables reasoning over relationships
- Retrieves both conceptual and exact information
- Reduces hallucinations in language models
- Works effectively with large enterprise knowledge bases